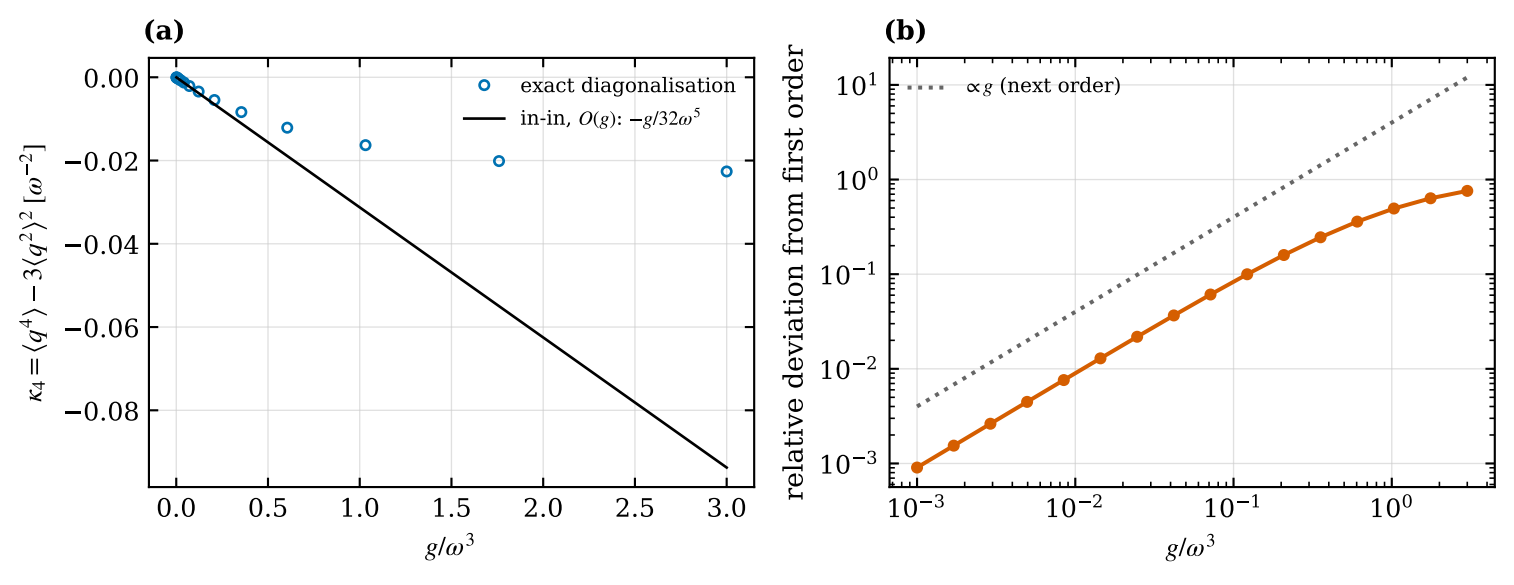
(a)
exact diagonalisation
in-in, $O(g)$: $-g/32\omega^5$
$\kappa_4 = \langle q^4 \rangle - 3\langle q^2 \rangle^2$ [ω^{-2}]
g/ω^3
(b)
$\propto g$ (next order)
relative deviation from first order
g/ω^3